Power input (5 V + PSU header)

VIN_SENSED → VIN_SENSE

File: power-input.kicad_sch

Opto inputs (817 module 2ch)

ALT_IN_ND → ALT_IN_N

PULSE_IN_ND → PULSE_IN_N

File: opto-inputs.kicad_sch

RS485 (MAX485 module)

RS485_RXD → RS485_RX

RS485_DE ← RS485_DE

RS485_TX ← RS485_TX

File: rs485.kicad_sch

ESP32 core + USB (DevKitC)

RS485_TX ← RS485_TX

EXP_IO5 → EXP_IO5

RTC_INT → RTC_INT

IMU_INT → IMU_INT

EXP_IO15 → EXP_IO15

UART2_RX → UART2_RX

UART2_TX ← UART2_TX

EXP_IO18 → EXP_IO18

EXP_IO19 → EXP_IO19

I2C_SDA → I2C_SDA

I2C_SCL → I2C_SCL

EXP_IO23 → EXP_IO23

RS485_DE ← RS485_DE

PULSE_IN_N → PULSE_IN_N

ALT_IN_N → ALT_IN_N

GPS_RXD ← GPS_RXD

EXP_IO33 → EXP_IO33

VIN_SENSE → VIN_SENSE

RS485_RX → RS485_RX

GPS_TXD → GPS_TXD

GPS_PPS → GPS_PPS

File: esp32-core.kicad_sch

RTC (DS3231 module)

I2C_SCL → I2C_SCL

I2C_SDA → I2C_SDA

RTC_INT ← RTC_INT

File: rtc-ds3231.kicad_sch

EEPROM (AT24C256 module)

I2C_SDA → I2C_SDA

I2C_SCL → I2C_SCL

File: eeprom-at24c256.kicad_sch

IMU (MPU9250 module -> MPU-6500 + LIS3MDL)

I2C_SCL → I2C_SCL

I2C_SDA → I2C_SDA

IMU_INT ← IMU_INT

File: imu.kicad_sch

Ports (UART, GPS, Qwiic, expansion)

UART2_RX ← UART2_RX

UART2_TX → UART2_TX

GPS_TXD ← GPS_TXD

GPS_RXD → GPS_RXD

GPS_PPS ← GPS_PPS

I2C_SDA → I2C_SDA

I2C_SCL → I2C_SCL

EXP_IO5 → EXP_IO5

EXP_IO19 → EXP_IO19

EXP_IO15 → EXP_IO15

EXP_IO18 → EXP_IO18

EXP_IO23 → EXP_IO23

EXP_IO33 → EXP_IO33

File: ports.kicad_sch

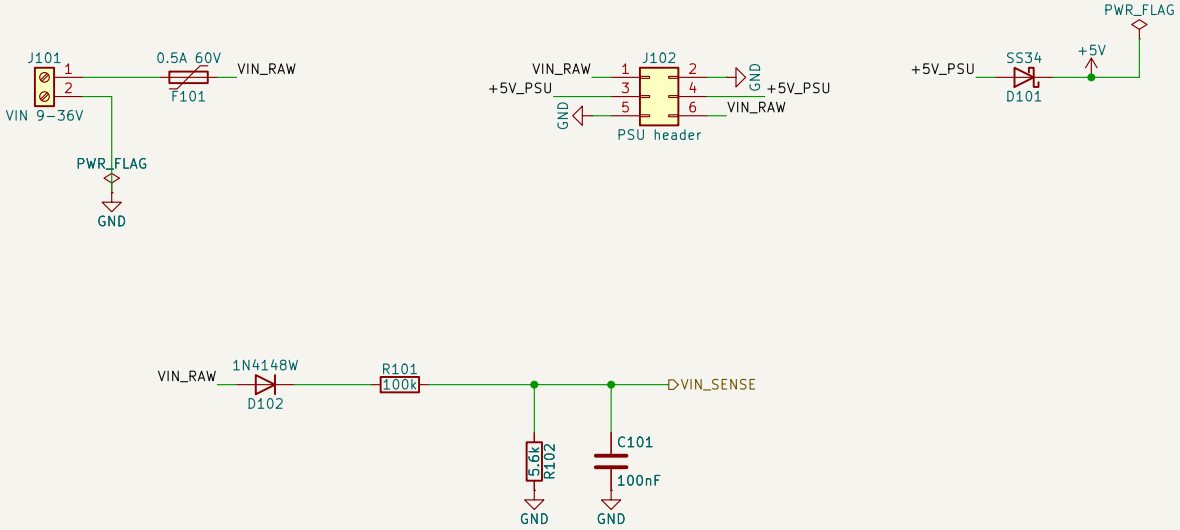
HOUR METER main board rev B (chip-down, JLCPCB in-stock parts)
One sheet per former module: Power input, Opto inputs (817 2ch), RS485 (MAX485), ESP32 core + USB (DevKitC),
RTC (DS3231), EEPROM (AT24C256, DIP socket), IMU (MPU9250 -> MPU-6500 + LIS3MDL), Ports (UART, GPS, Qwiic, expansion).
Sheet pins join by net name (labels at every pin). +5V, +3V3 and GND are global power symbols.
The board runs from 5 V; the 9-36 V supply is the plug-on PSU-B board (schematic/hour-meter-psu-b) on J102.
Reference designators are numbered per sheet: 1xx power, 2xx core, 3xx opto, 4xx RTC, 5xx EEPROM, 6xx IMU, 7xx RS485, 8xx ports.

Sheet: /
File: hour-meter-board.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/9

POWER INPUT
The main board runs from 5 V. The 9–36 V -> 5.3 V converter is a plug-on PSU board (hour-meter-psu-b) on J102;
the vehicle wire stays on J101. Alternative: feed 5.3 V from an LM2596 module into J102 pins 3/4 (GND on 2/5).
J102 pinout is 180-degree safe: 1 VIN, 2 GND, 3 +5V_PSU, 4 +5V_PSU, 5 GND, 6 VIN.
USB VBUS is OR-ed into +5V on the esp32-core sheet. VIN_SENSE = VIN x 5.6/105.6 (GPIO34, ADC1_CH6).
H101 = PSU standoff; H102–H105 = board corner holes.



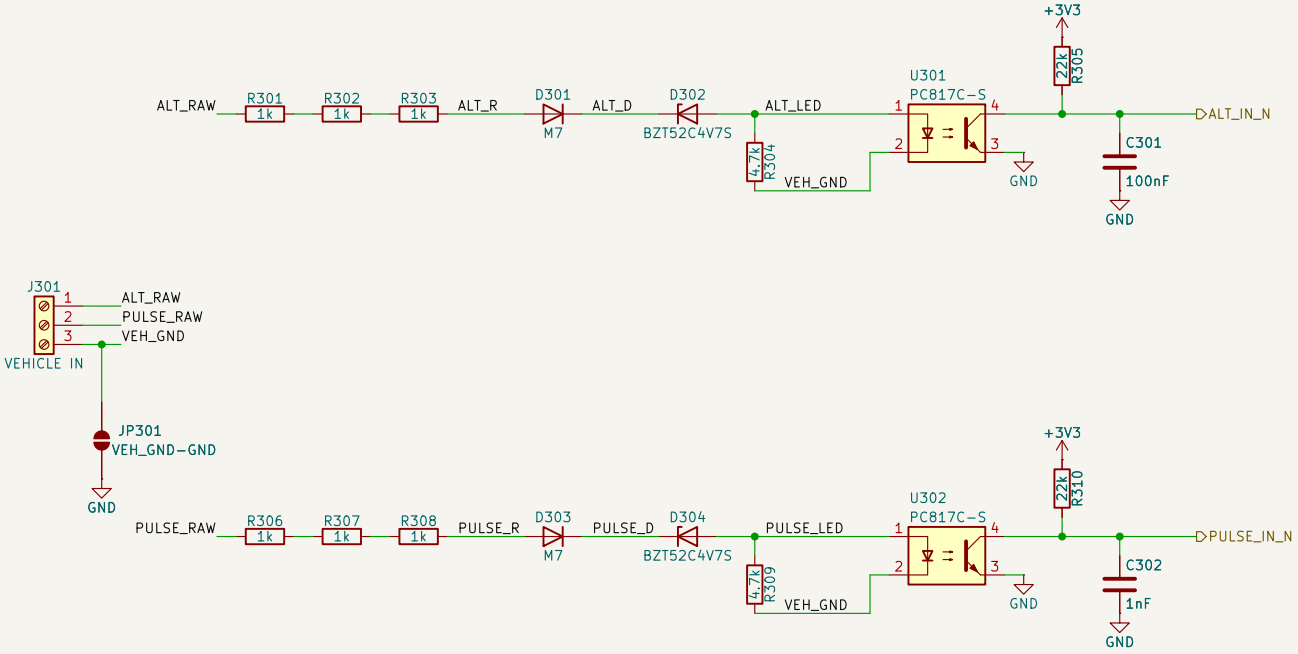
H101 PSU standoff M3 H102 M3 H103 M3 H104 M3 H105 M3

Sheet: /Power input (5 V + PSU header)/
File: power-input.kicad_sch

Title:

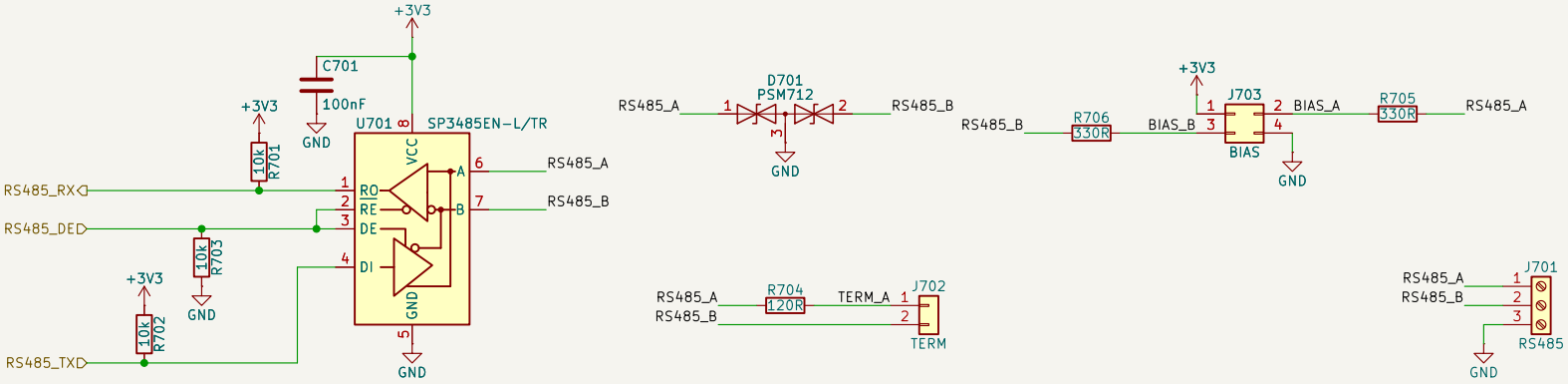
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/9

OPTO INPUTS (replaces 817 module 2ch), redesigned for 9–36 V vehicle signals
Each channel: 3 x 1k 1206 in series (<= 0.1 W each) -> M7 reverse block -> BZT52C4V7S 4.7 V zener -> PC817C LED, 4.7k bleed across the LED.
Threshold ~7.3 V (4.7 V zener + LED + diode + bleed). LED current >= 0.55 mA at 9 V and 9.8 mA at 36 V.
ALT_IN_N / PULSE_IN_N: open collector, 22k pull-up to +3V3, active LOW (firmware altActiveHigh=false).
ALT filter 22k x 100nF = 2.2 ms; PULSE filter 22k x 1nF = 22 us.
Wire ALT to the alternator L/D+ terminal or the oil-pressure switch (see esp32/docs/WIRING.md).
JP301 open (default) keeps the vehicle side isolated: VEH_GND (J301 pin 3) is separate from GND.



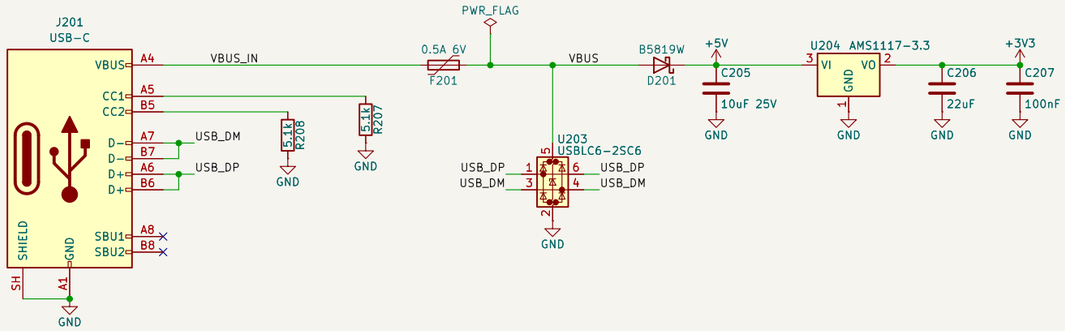
Sheet: /Opto inputs (817 module 2ch)/		
File: opto-inputs.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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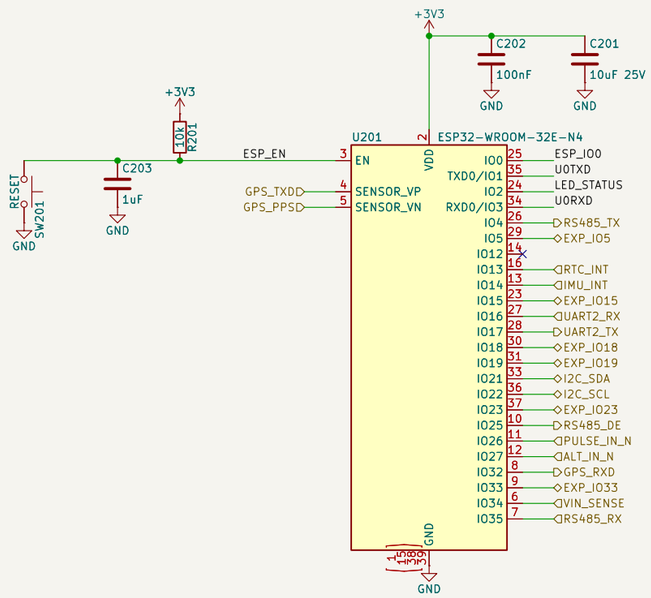
RS485 (replaces MAX485 module)
U701 SP3485EN 3.3 V half-duplex transceiver (JLC basic part) on UART1: GPIO4 = TX (DI), GPIO35 = RX (RO),
GPIO25 = DE + /RE (ESP-IDF UART RS485 half-duplex mode).
R701 / R702 hold RX / TX high (RO floats while transmitting); R703 holds DE low (receive) while the MCU is in reset.
D701 PSM712 clamps A and B to GND (asymmetric −7 V / +12 V, the RS485 common-mode range).
J702 TERM: fit a shunt (120R across A–B) only on the two nodes at the ends of the bus.
J703 BIAS: fit both shunts, across 1–2 and 3–4 (A to 3V3 and B to GND via 330R), on exactly one node:
idle A–B = 3.3 V x 60/(330+330+60) = 0.28 V with both ends terminated (receiver threshold 0.2 V).
Jumpers off = no termination and no bias.

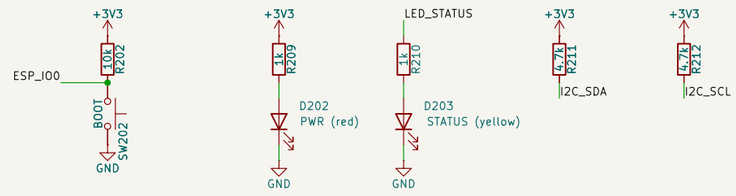


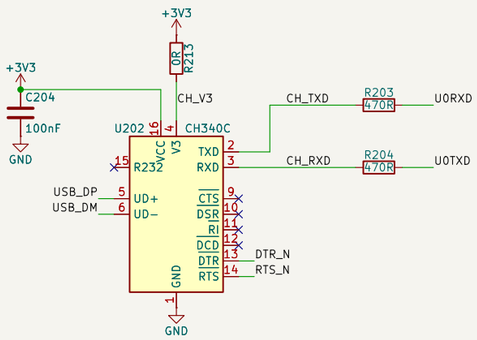
Sheet: /RS485 (MAX485 module)/		
File: rs485.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/9

ESP32 CORE + USB (replaces ESP32-DevKitC)
Chip-down DevKitC V4: ESP32-WROOM-32E-N4, USB-C (5.1k CC pull-downs, USBLC6 ESD), CH340C instead of CP2102N.
USB VBUS -> 0.5 A polyfuse -> B5819W -> +5V (OR-ed with the PSU feed from power-input); AMS1117-3.3 -> +3V3.
CH340C runs at 3.3 V: VCC on +3V3, V3 tied to +3V3 through 0R R213, so its TXD/RXD are 3.3 V logic; 470R in series on both UART0 lines.
Auto-program (esptool): DTR#=1, RTS#=0 -> EN low = reset; DTR#=0, RTS#=1 -> IO0 low = boot; DTR#=RTS# -> both released.
I2C pull-ups (4.7k to +3V3) live only on this sheet; module sheets add none.
GPIO12 (MTDI strap) must stay unconnected. GPIO34/35/36/39 are input-only (no internal pull-ups).





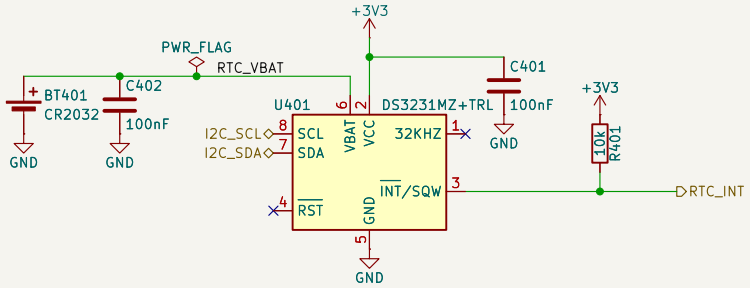






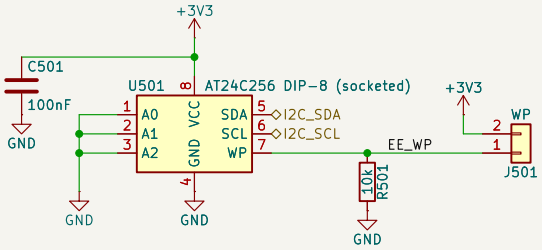
Sheet: /ESP32 core + USB (DevKitC)/		
File: esp32-core.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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RTC (replaces DS3231 ZS-042 module)
U401 DS3231MZ+TRL, I2C address 0x68, +/-5 ppm (SOIC-8). Register-compatible upgrade: DS3231SN#T&R (LCSC C9866, SOIC-16, +/-2 ppm).
BT401 CR2032 is a primary cell: no charge path (the ZS-042's 200R + 1N4148 charger is deliberately removed).
The module's AT24C32 EEPROM (0x57) is not fitted. I2C pull-ups are on esp32-core.
INT/SQW is open drain: R401 10k pull-up to +3V3 -> RTC_INT. 32KHZ and RST are not used.
PCB: the battery holder is nickel-plated steel, keep it >= 20 mm from the magnetometer.



Sheet: /RTC (DS3231 module)/		
File: rtc-ds3231.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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EEPROM (replaces AT24C256 module)
U501 is a swappable chip in a DIP-8 socket (only the socket is assembled): the user inserts an AT24C256 DIP-8,
e.g. LCSC C434602 AT24C256N or C411624 24LC256-I/P.
I2C address 0x50 (A0 = A1 = A2 = GND). The firmware layout is fixed at 32 KB (EepromLayout.h), so fit a 24C256 or 24C512.
WP: R501 10k pulls EE_WP low (writes enabled); a jumper on J501 ties WP to +3V3 and write-protects the chip.
No I2C pull-ups here (they are on esp32-core).

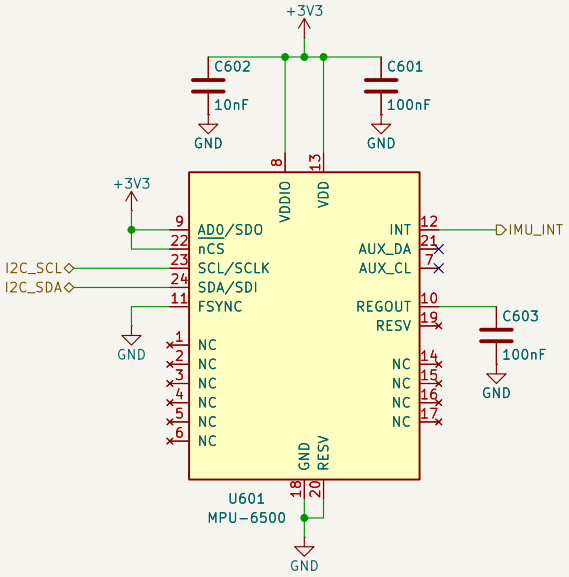


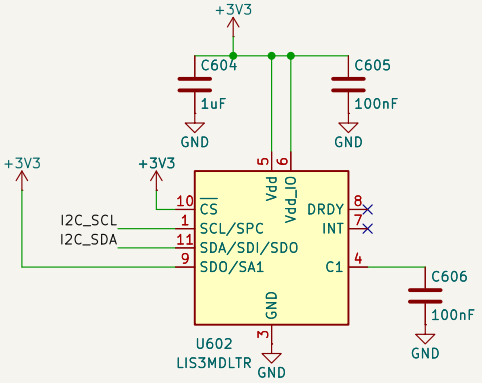
Sheet: /EEPROM (AT24C256 module)/
File: eeprom-at24c256.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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IMU (replaces MPU9250 GY-9250 module)
MPU-9250 = MPU-6500 die + AK8963 magnetometer and is out of stock at JLCPCB, so the two functions are now separate chips:
U601 MPU-6500 accel/gyro at I2C 0x69 (AD0 = 3V3; 0x68 is the DS3231). Firmware Mpu9250.cpp already accepts WHO_AM_I 0x70.
U602 LIS3MDL magnetometer at I2C 0x1E (SA1 = 3V3); needs a new small firmware driver (it replaces the AK8963 at 0x0C).
Both chips in I2C mode (CS = 3V3). No LDO: both run from the board 3V3. No I2C pull-ups here (one 4.7k pair on esp32-core).
MPU-6500: VDD 100nF, VDDIO 10nF, REGOUT 100nF; FSYNC and RESV pin 20 to GND; RESV pin 19 and NC pins left open.
PCB: place both chips with the same orientation; keep the LIS3MDL at least 20 mm from the PSU header,
the CR2032 holder and the screw terminals.



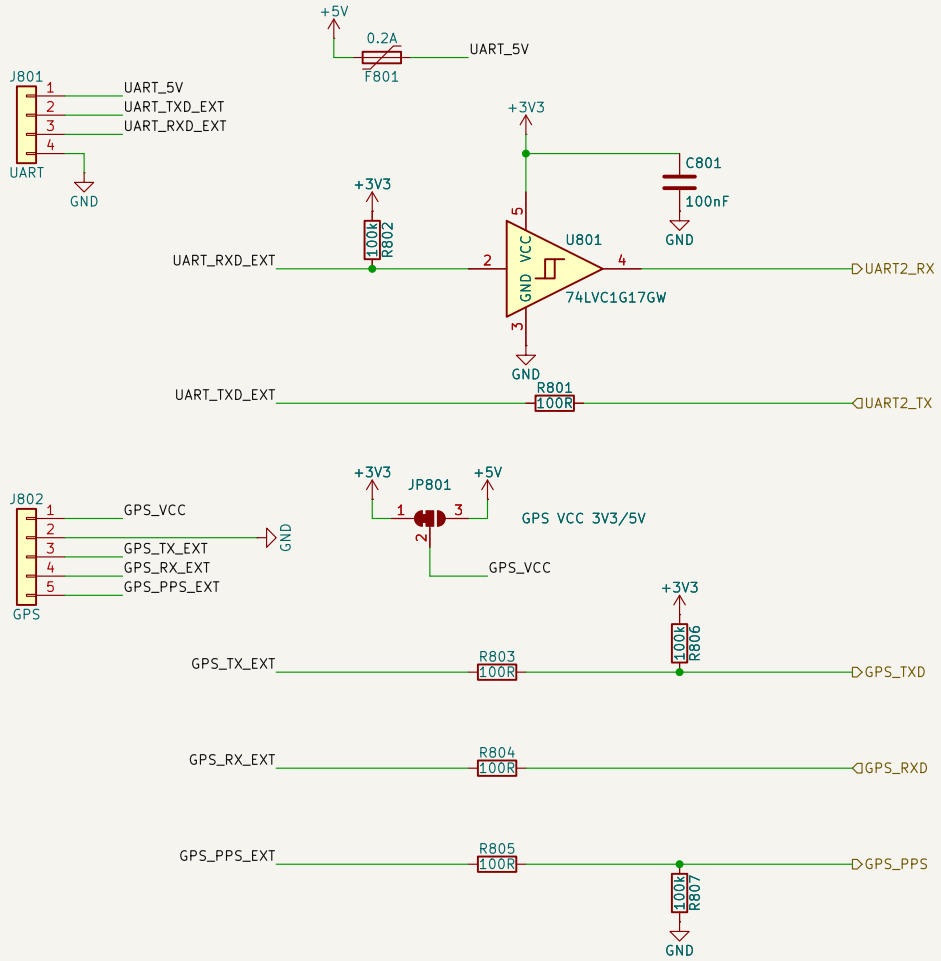


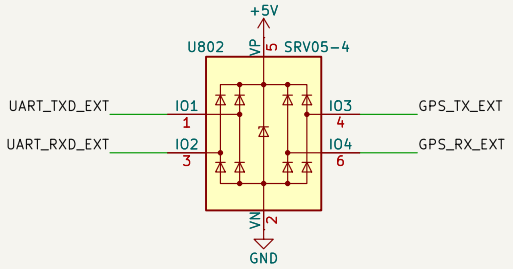
Sheet: /IMU (MPU9250 module -> MPU-6500 + LIS3MDL)/
File: imu.kicad_sch

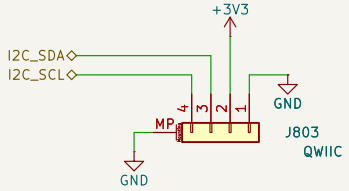
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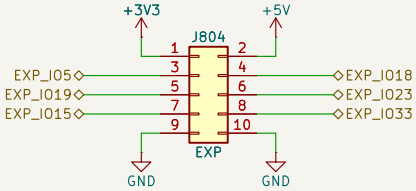
Size: A4	Date:	**Rev:**
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PORTS
J801 UART sensor port: UART2 = GPIO17 TX (3.3 V out via R801 100R) / GPIO16 RX via U801 74LVC1G17 (5 V–tolerant input, R802 keeps it idle–high); pin 1 = 5 V, 200 mA via F801 polyfuse. U802 SRV05–4 = ESD on the four external UART/GPS lines.
J802 GPS card header for u–blox type modules: UART on GPIO36 (GPS TX) / GPIO32 (GPS RX), PPS on GPIO39, 100R series. R806 holds GPS_TXD high with no module fitted (GPIO36 is input–only, no internal pull–up); R807 holds PPS low.
GPS supply = 3V3 by default (JP801 bridged 1–2); cut 1–2 and bridge 2–3 for a 5 V module.
u–blox modules with I2C can use the Qwiic port J803 (address 0x42) instead.
J804 expansion: IO5 and IO15 are strapping pins – do not pull them low at boot.









Sheet: /Ports (UART, GPS, Qwiic, expansion)/
File: ports.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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